

Functional Annotation Report: tauC (PP_0231, UniProt Q88RA2)

Taurine/alkanesulfonate ABC transporter permease subunit — *Pseudomonas putida* KT2440

1. Summary (Answer to the Research Question)

tauC (locus *PP_0231*; UniProt **Q88RA2**) of *Pseudomonas putida* KT2440 encodes the **integral inner-membrane permease (transmembrane) subunit of the taurine/alkanesulfonate ABC importer TauABC**. Its primary function is **membrane transport, not catalysis**: TauC forms the substrate-translocation channel through which **taurine (2-aminoethanesulfonate)** and related aliphatic (alkane)sulfonates are pumped from the periplasm into the cytoplasm, driven by ATP hydrolysis at the associated ABC ATPase subunit TauB. The physiological purpose is **sulfur scavenging**: under sulfate/cysteine starvation the cell imports organosulfonates and releases their sulfur intracellularly (via the α -ketoglutarate-dependent taurine dioxygenase TauD), feeding the sulfur-assimilation (cys) pathway.

Note on the EC number. UniProt lists "EC 3.6.3.36" (taurine-transporting ATPase) for the whole transport system. This ATP-hydrolysis activity resides in the **TauB** nucleotide-binding subunit; **TauC itself is the non-catalytic membrane channel**. Assigning the EC to the permease is a system-level annotation, not evidence that TauC hydrolyzes ATP.

2. Gene/Protein Identity — Verification

Attribute	Value	Consistency check
UniProt	Q88RA2	✓
Gene / locus	<i>tauC</i> / PP_0231	✓ matches <i>tau</i> operon nomenclature
Organism	<i>Pseudomonas putida</i> KT2440	✓
Description	Taurine ABC transporter permease subunit	✓
Family	Binding-protein-dependent transport system	✓ ABC importer
Domains	MetI-like (IPR000515; IPR035906); Pfam PF00528 (BPD_transp_1)	✓ canonical ABC-importer permease fold

Identity is confirmed. The MetI-like transmembrane fold (PF00528, "binding-protein-dependent transport, inner membrane component") is the hallmark of ABC-importer permeases, and the *tauC* symbol/organism/family/domain set is internally consistent with the well-characterized bacterial **tauABCD** taurine-uptake system. The literature cited below describes this exact system (in *E. coli* as the paradigm, with the *Pseudomonas putida* sulfur regulon explicitly addressed by Kertesz 2000).

3. Molecular Function — What TauC Does

3.1 A subunit of a multi-component ABC importer

Bacterial organosulfonate uptake is mediated by **binding-protein-dependent (ABC-type) transporters**. In the paradigmatic *E. coli* **tauABCD** operon the division of labor is:

- **TauA** — periplasmic **substrate-binding protein (SBP)**; captures taurine and delivers it to the membrane complex.
- **TauC** — **transmembrane permease** (MetI-like); forms the translocation pathway across the inner membrane. (*This protein, Q88RA2.*)
- **TauB** — cytoplasmic **ABC ATPase** (nucleotide-binding domain); ATP binding/hydrolysis powers the conformational cycle.

- **TauD** — cytoplasmic **desulfonation enzyme** (not part of the transporter; processes the imported substrate).

"The amino acid sequences of SsuABC and TauABC exhibit similarity to those of components of the ATP-binding cassette transporter superfamily, suggesting that two uptake systems for alkanesulfonates are present in *E. coli*." — Eichhorn, van der Ploeg & Leisinger, 2000

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Because TauC lacks a periplasmic capture domain and lacks the ABC nucleotide-binding cassette, it functions **only in complex** — as the membrane channel that receives substrate from TauA and is gated by conformational signals from TauB. Deletion analysis in *E. coli* showed that **all three of TauA, TauB and TauC are required** for taurine uptake as a sulfur source, and that non-productive "hybrid" transporters cannot support growth on sulfonates (Eichhorn et al. 2000,

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3.2 Substrate and specificity

The transported substrate is **taurine (2-aminoethanesulfonate)** and, more broadly, **aliphatic/alkanesulfonates**. Substrate selectivity is largely set by the **periplasmic binding protein (TauA)** rather than by the permease:

"TauA and SsuA are substrate-binding proteins (SBPs) that bind and bring the alkanesulfonates to the ABC importer for transport... the different levels of hydration of the binding site contributed to the selectivity for taurine over the other alkanesulfonates."

— Qu et al., 2019

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Thus the TauABC system is characterized as a **taurine-preferring** importer, while the paralogous **SsuABC** system handles a broader range of alkanesulfonates; the two systems overlap but are not redundant (Eichhorn et al. 2000,

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 van der Ploeg et al. 2001,
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3.3 Strain-specific genomic evidence (KT2440)

Genomic analysis of the KT2440 chromosome confirms that PP_0231 is embedded in a **complete, co-directional (complement-strand) taurine transport-and-desulfonation operon** — direct strain-specific evidence rather than *E. coli* homology alone:

Locus	Gene	Product	KO / EC	Position
PP_0230	<i>atsK</i>	α -ketoglutarate-dependent taurine/sulfonate dioxygenase	K03119 / EC 1.14.11.17	284616–285449
PP_0231	<i>tauC</i>	ABC transporter permease (<i>this gene</i>)	K15552	285498–286337
PP_0232	<i>tauB-II</i>	ABC transporter ATP-binding subunit	K10831 / EC 7.6.2.7	286334–287122
PP_0233	<i>tauA</i>	periplasmic substrate-binding protein	K15551	287136–288107

All four genes are contiguous and on the same strand, forming the functional module: **capture (TauA) → translocation (TauC) → energization (TauB) → intracellular desulfonation (AtsK/TauD)**. KEGG assigns PP_0231 to **Sulfur metabolism (ppu00920)** and **ABC transporters (ppu02010)**. Notably, the modern EC number for the ATP-hydrolysis step, **EC 7.6.2.7 (taurine-transporting ATPase)**, is annotated to **TauB (PP_0232)** — directly confirming that catalysis resides in the ATPase, and that the "Hydrolase/EC 3.6.3.36" tag on TauC is a spurious carry-over onto the non-catalytic permease.

Protein-level features (UniProt Q88RA2): 279 aa, 29.9 kDa, **multi-pass inner-membrane protein with 6 predicted transmembrane α -helices** (res. 30–52, 91–113, 125–144, 150–169, 213–234, 240–260) and an ABC transmembrane type-1 domain (PROSITE PS50928, res. 84–264). Family/clustering evidence: eggNOG **COG0600**, CDD **TM_PBP2**, PANTHER **PTHR30151:SF25 "Taurine transport system permease protein TauC"**. GO annotations: *alkanesulfonate transmembrane transport* (GO:0042918), *cellular response to sulfur starvation* (GO:0010438), *plasma membrane* (GO:0005886).

3.4 Orthology to experimentally characterized *E. coli* TauC

A global (Needleman–Wunsch) alignment computed here shows that *P. putida* TauC (Q88RA2, 279 aa) is **64.7% identical (176/272 aligned residues, near full-length coverage)** to *E. coli* K-12 **TauC (Q47539, "Taurine transport system permease protein TauC," 275 aa)**. This far exceeds the ~30–40% threshold used to infer orthology for membrane transporters (unrelated *E. coli* permeases score much lower, e.g. PerM 36.5%). Such high, full-length identity establishes Q88RA2 as a **true ortholog** of the experimentally defined *E. coli* taurine permease — justifying direct transfer of the substrate specificity, subunit-cooperation, mechanism, and physiological-role data below to the *P. putida* protein.

3.5 Mechanism (structural/evolutionary inference)

The **MetI-like** annotation places TauC in the same mechanistic class as the *E. coli* methionine-importer permease MetI. These importers operate by an **alternating-access** mechanism: two permease subunits form a translocation cavity that switches between outward- and inward-facing states, driven by the ATP-hydrolysis cycle of the ABC ATPase subunits. This is a **structure/evolution-based inference** from the conserved MetI-like fold (IPR000515/IPR035906, PF00528) rather than a direct experimental determination for Q88RA2 itself.

4. Subcellular Localization — Where It Acts

TauC is an **integral protein of the cytoplasmic (inner) membrane** (UniProt: "Cell membrane; Multi-pass membrane protein"; GO:0005886 plasma membrane). Consistent with the MetI-like fold, it comprises **six predicted membrane-spanning α -helices** that build the substrate channel. Its two functional interfaces are: - **Periplasmic face** — docks the substrate-loaded periplasmic binding protein TauA; - **Cytoplasmic face** — couples to the ABC ATPase TauB.

Transport therefore occurs **across the inner membrane** (periplasm → cytoplasm), and the downstream desulfonation of the imported taurine takes place **in the cytoplasm** (TauD).

5. Biological Process / Pathway Context

5.1 Organosulfur scavenging under sulfur limitation

In aerobic soil bacteria such as *Pseudomonas*, **sulfonates and sulfate esters constitute >95% of soil sulfur**, and organisms scavenge this sulfur when inorganic sulfate/cysteine are unavailable (Kertesz 2000,

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TauABC is the **uptake gateway** for this pathway:

1. **Uptake** — TauA (periplasm) → **TauC permease** → cytoplasm, energized by TauB (ATP).
(*TauC step*.)
2. **Desulfonation** — cytoplasmic **TauD**, an Fe(II)/ α -ketoglutarate-dependent dioxygenase, "preferentially liberates sulfite from taurine (2-aminoethanesulfonic acid)" (van der Ploeg et al. 2001,

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3. **Assimilation** — released sulfite/sulfur enters cysteine/methionine biosynthesis.

Note: An SsuD-dependent, TauD-independent route for taurine sulfur assimilation also exists (Nishikawa et al. 2018,

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underscoring that the **transport step (TauABC)** and the **desulfonation step are genetically separable**; TauC's role is strictly transport.

5.2 Regulation

The *tau* genes are part of the **extended cys (sulfur-starvation) regulon**: they are **expressed only under sulfate/cysteine starvation** and repressed by sulfate. Control is exerted by the LysR-type regulators **CysB (master regulator)** and **Cbl** in *E. coli*:

"Starvation for sulfate leads to the expression of the tauABCD and ssuEADCB genes... Expression of the tau and ssu genes requires the LysR-type transcriptional regulatory proteins CysB and Cbl." — van der Ploeg et al., 2001

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In *Pseudomonas putida* specifically, an **additional regulator, AsfR**, is required for sulfonate desulfonation (Kertesz 2000,

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placing PP_0231 within a *Pseudomonas*-specific sulfur-limitation regulatory network. The re-annotated KT2440 genome (Belda et al. 2016,

[P 26913973](#))

affirms the breadth of catabolic/nutrient-scavenging capacity encoded in this strain.

6. Evidence Summary

Claim	Evidence type	Source
tauC is the transmembrane permease of an ABC importer	Sequence/domain annotation (MetI-like, PF00528); operon architecture	UniProt Q88RA2; InterPro; Eichhorn 2000 (10781534)
Q88RA2 is a true ortholog of <i>E. coli</i> TauC	Global alignment: 64.7% identity, full-length	This work vs UniProt Q47539
Complete tau operon (<i>atsK</i> – <i>tauC</i> – <i>tauB</i> – <i>tauA</i>) in KT2440	Genome/KEGG neighborhood analysis	This work; KEGG ppu:PP_0230–0233
Substrate = taurine / alkanesulfonates	Genetic deletion + transport phenotypes; SBP binding thermodynamics	Eichhorn 2000 (10781534); Qu 2019 (31802112)
All of TauA/B/C required for uptake	In-frame deletion & complementation	Eichhorn 2000 (10781534)
Specificity set by binding protein TauA	Crystal structure + ITC + MD	Qu 2019 (31802112)
Function = sulfur scavenging under sulfate starvation	Expression/regulation studies	van der Ploeg 2001 (11479697); Kertesz 2000 (10717312)
Downstream desulfonation by TauD (α -KG dioxygenase)	Enzymology / genetics	van der Ploeg 2001 (11479697); Nishikawa 2018 (30277858)
<i>P. putida</i> -specific regulator AsfR	Review of <i>Pseudomonas</i> sulfur metabolism	Kertesz 2000 (10717312)
Mechanism (alternating access)	Structural inference from MetI-like fold	InterPro IPR000515/ IPR035906

Evidence caveat: Direct biochemical/structural characterization of the *P. putida* KT2440 TauC protein (Q88RA2) itself was not found; the functional assignment rests on (i) unambiguous domain/family membership and a 6-TM MetI-like topology, (ii) **strain-specific operon context in KT2440** (contiguous *atsK*–*tauC*–*tauB*–*tauA* cluster, verified here from the genome/KEGG),

and (iii) detailed experimental work on the orthologous *E. coli* TauABC system and the *Pseudomonas* sulfur regulon. This is a high-confidence annotation supported by convergent bioinformatic, genomic, and comparative-experimental evidence.

7. Supported vs. Refuted Hypotheses

Supported - TauC is the membrane permease subunit of an ABC-type importer (not an enzyme). ✓ - The imported substrate is taurine/alkanesulfonates. ✓ - It localizes to the inner (cytoplasmic) membrane. ✓ - It functions in sulfur scavenging under sulfate/cysteine limitation, within the CysB/Cbl (AsfR in *Pseudomonas*) regulon. ✓

Refuted / corrected - That TauC itself catalyzes ATP hydrolysis (EC 3.6.3.36): the ATPase activity belongs to TauB; the EC is a system-level label. ✗ for the permease. - That taurine sulfur assimilation strictly requires TauD downstream: an SsuD-dependent bypass exists (Nishikawa 2018), though this does not change TauC's transport role. ✗ (as an obligatory single route)

8. Limitations and Future Directions

- **No direct structural or biochemical data** exist for Q88RA2 specifically; an experimental structure (e.g., cryo-EM of the *P. putida* TauB₂C₂ complex) or an AlphaFold model would confirm the predicted membrane topology and TauA/TauB interfaces.
 - **Substrate range in *P. putida*** (beyond taurine) has not been directly measured for this transporter; transport assays with defined alkanesulfonates would resolve overlap with the SsuABC system.
 - **Regulatory wiring** (CysB/Cbl/AsfR control of PP_0231) is inferred from *E. coli* and general *Pseudomonas* studies; strain-specific transcriptomics under sulfate starvation would confirm induction.
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References (PMIDs)

- Eichhorn, van der Ploeg & Leisinger, 2000 — Deletion analysis of *E. coli* taurine and alkanesulfonate transport (10781534)
- van der Ploeg, Eichhorn & Leisinger, 2001 — Sulfonate-sulfur metabolism and its regulation in *E. coli* (11479697)
- Qu, ElOmari, Wagner, De Simone & Beis, 2019 — TauA desolvation dictates ligand specificity of TauABC (31802112)
- Kertesz, 2000 — Sulfonate/sulfate ester metabolism in gram-negative bacteria (incl. *P. putida* AsfR) (10717312)
- Nishikawa, Shen & Ogawa, 2018 — TauD-independent taurine assimilation in *E. coli* (30277858)
- Belda et al., 2016 — Revisited genome of *P. putida* KT2440 (26913973)

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